

NaileR, a game changer for interpreting
continuous and categorical latent variables
with an AI twist: an application to consumer
and sensory data

*“Prep, polish, top coat, this vanity case is exactly what you need to
put the final touch to your statistical analysis”*

Sébastien Lê



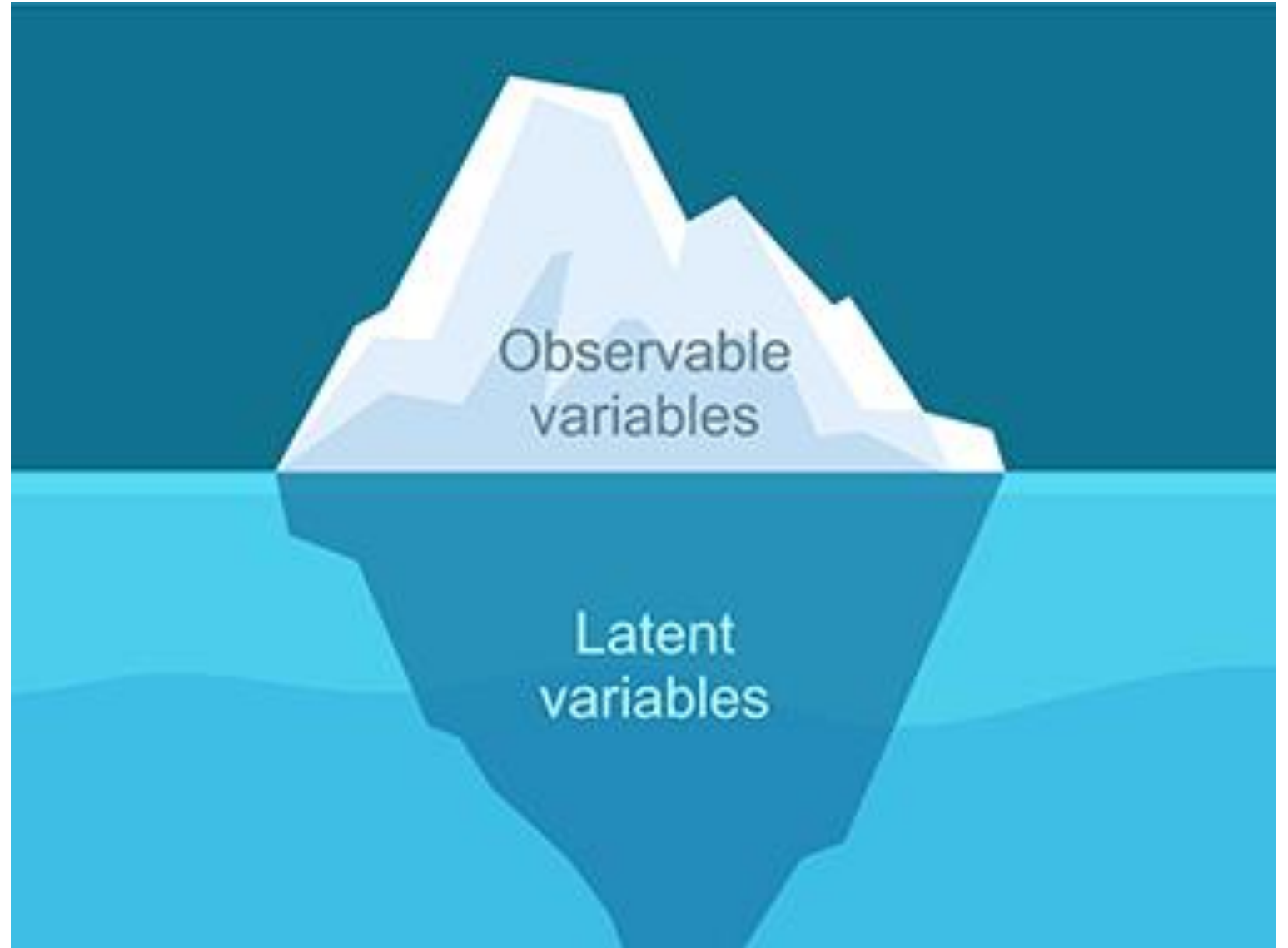
continuous
variable:
“5-point
emotional
scale”

5	<u>I can't stand this and ready to explode.</u> I want to hit or kick someone, or throw something. I need help from an adult. OUT OF CONTROL	
4	<u>I am getting too angry.</u> My face feels hot. I need to go to my safe spot to calm down. ANGRY	
3	<u>I am getting frustrated.</u> I will tell an adult. Take deep breaths. FRUSTRATED	
2	<u>I am doing OK.</u> I am not happy. I can stay where I am and keep working. I can control myself. ANNOYED	
1	<u>I am doing great.</u> I feel good about myself and about what is going on around me. HAPPY	

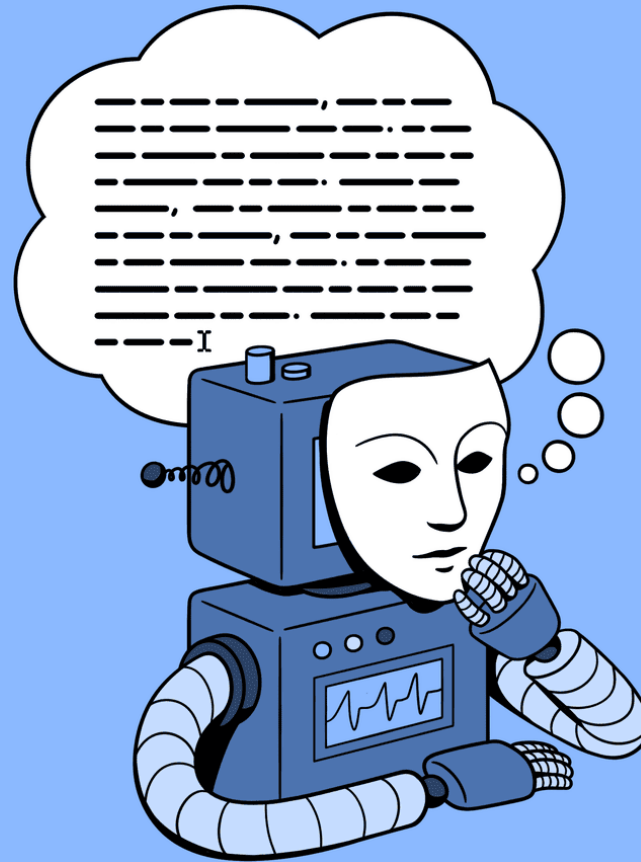


categorical
variable:
“warm/cool”

latent
variable:
“that cannot
be directly
observed”



AI twist:
“large
language
model”

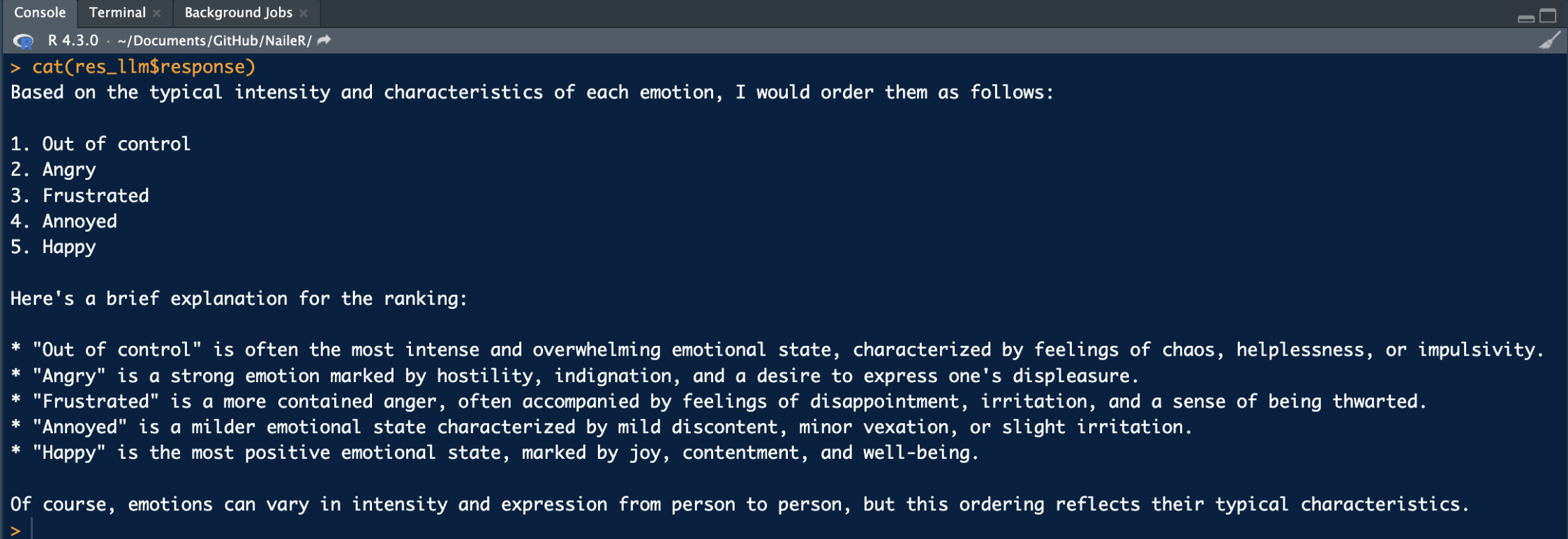


Large Language Model (LLM)

['lärj 'laŋ-gwij 'mä-dəl]

A deep learning algorithm that's equipped to summarize, translate, predict, and generate human-sounding text to convey ideas and concepts.

“Please order these emotions according to a scale: frustrated, angry, happy, out of control, annoyed”



```
Console Terminal Background Jobs
R 4.3.0 · ~/Documents/GitHub/NaileR/
> cat(res_llm$response)
Based on the typical intensity and characteristics of each emotion, I would order them as follows:

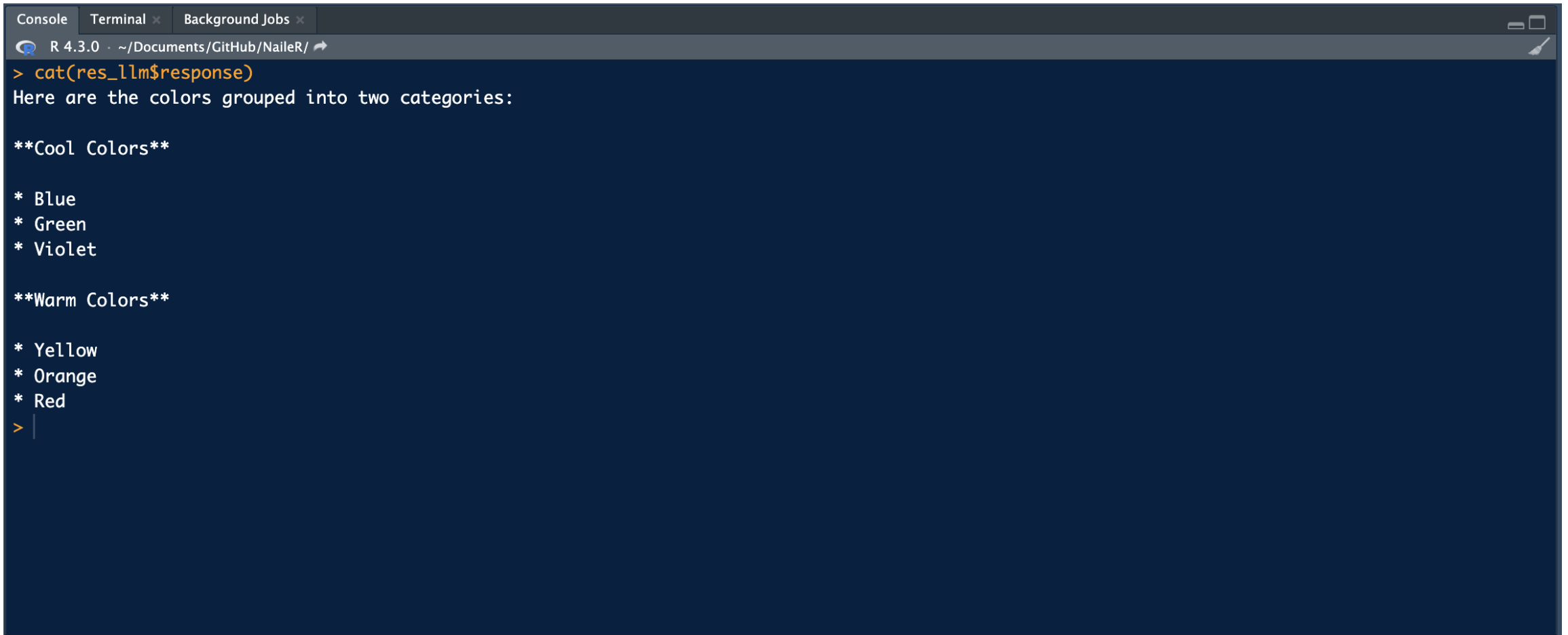
1. Out of control
2. Angry
3. Frustrated
4. Annoyed
5. Happy

Here's a brief explanation for the ranking:

* "Out of control" is often the most intense and overwhelming emotional state, characterized by feelings of chaos, helplessness, or impulsivity.
* "Angry" is a strong emotion marked by hostility, indignation, and a desire to express one's displeasure.
* "Frustrated" is a more contained anger, often accompanied by feelings of disappointment, irritation, and a sense of being thwarted.
* "Annoyed" is a milder emotional state characterized by mild discontent, minor vexation, or slight irritation.
* "Happy" is the most positive emotional state, marked by joy, contentment, and well-being.

Of course, emotions can vary in intensity and expression from person to person, but this ordering reflects their typical characteristics.
> |
```

“Please group these colors into two groups:
blue, yellow, green, orange, violet, red”



```
Console | Terminal x | Background Jobs x | R 4.3.0 · ~/Documents/GitHub/NaileR/ ↗
> cat(res_llm$response)
Here are the colors grouped into two categories:

**Cool Colors**

* Blue
* Green
* Violet

**Warm Colors**

* Yellow
* Orange
* Red
> |
```

“Please give a name to a scale to would sort emotions that way: happy, annoyed, frustrated, angry, out of control”

```
Console Terminal Background Jobs
R 4.3.0 · ~/Documents/GitHub/NaileR/
> cat(res_llm$response)
What a great idea!

After some thought, I'd like to propose the following name for this emotional scale:

**EmoSphere**

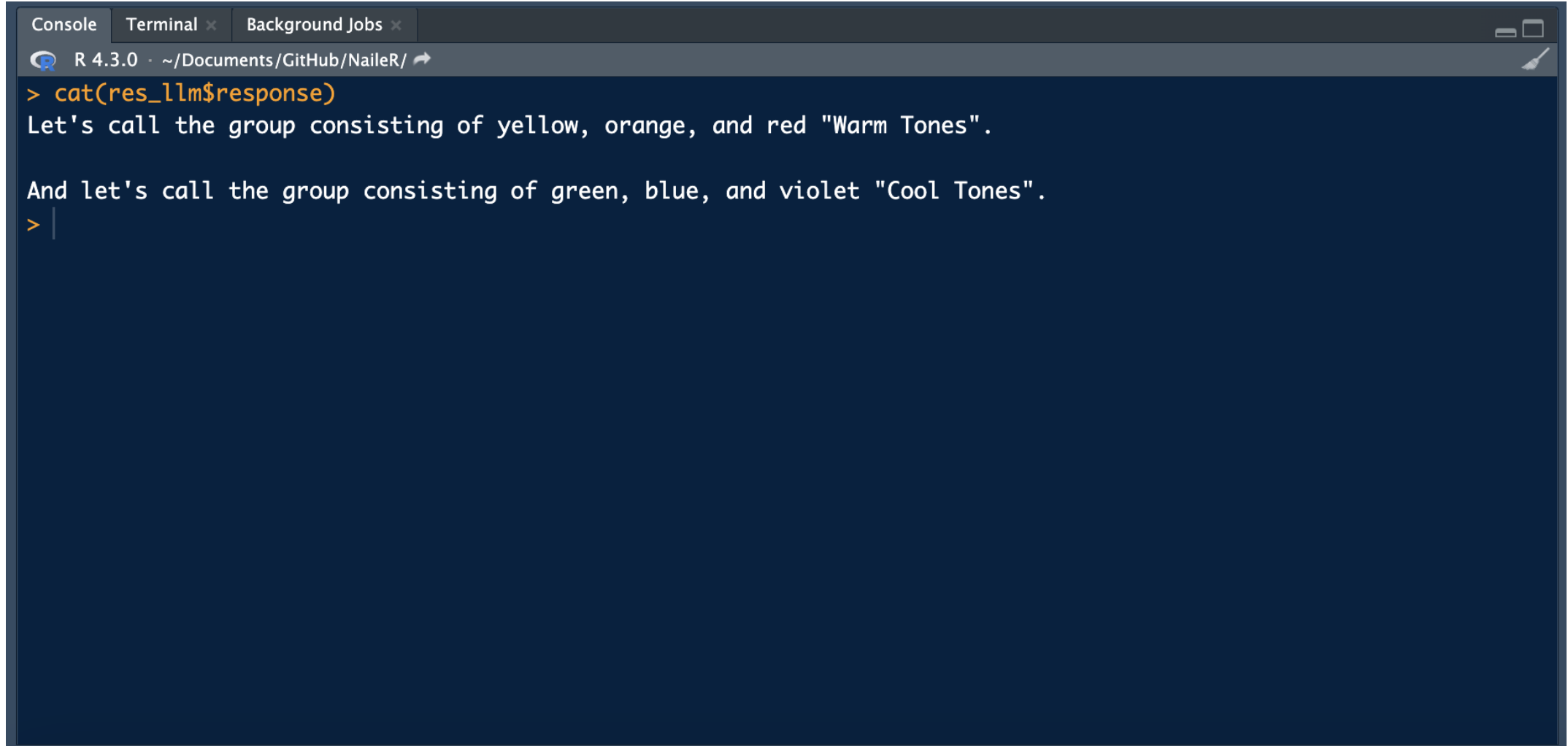
The EmoSphere scale categorizes emotions in a circular pattern, with "Happy" at one end and "Out of Control" at the other. Here's how it works:

1. **Happy**: The center point of the EmoSphere, where emotions are positive and balanced.
2. **Annoyed**: A slight shift to the left, indicating mild irritation or displeasure.
3. **Frustrated**: A moderate step up in intensity, representing feelings of discontent and agitation.
4. **Angry**: A more significant increase in emotional intensity, characterized by strong negative emotions like hostility or resentment.
5. **Out of Control**: The outermost point on the EmoSphere, where emotions are overwhelming and potentially explosive.

This scale acknowledges that emotions don't always follow a linear progression, but can ebb and flow in a circular pattern. By using the EmoSphere, individuals can better understand their emotional state and take steps to manage their feelings more effectively.

What do you think? Does this name resonate with you, or would you prefer something else?
> |
```


“Please give a name to each one of two groups:
yellow, orange, red, and green, blue, violet”



```
Console Terminal x Background Jobs x
R 4.3.0 · ~/Documents/GitHub/NaileR/
> cat(res_llm$response)
Let's call the group consisting of yellow, orange, and red "Warm Tones".

And let's call the group consisting of green, blue, and violet "Cool Tones".
> |
```

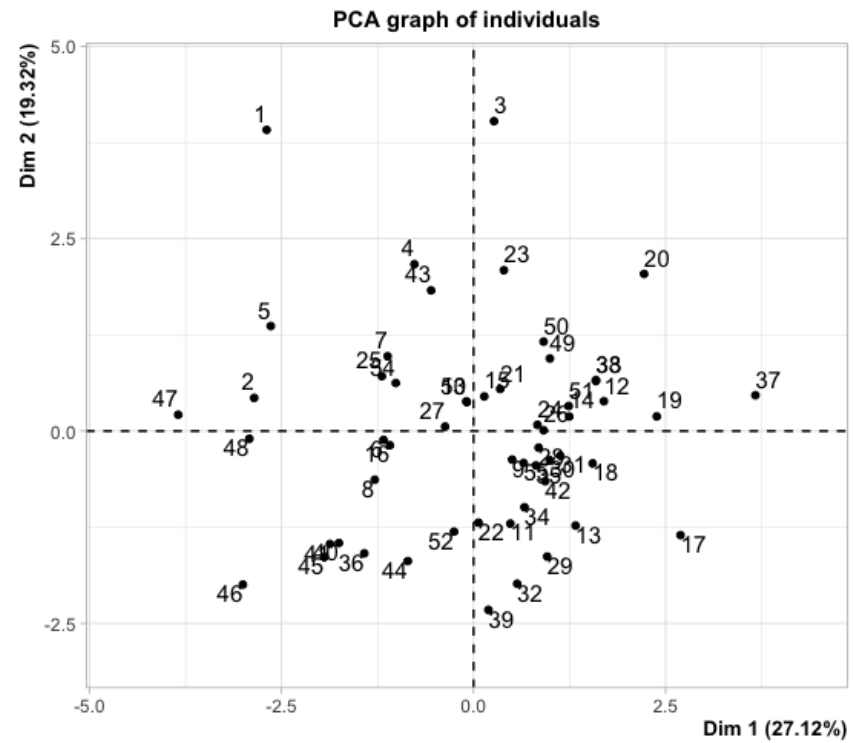
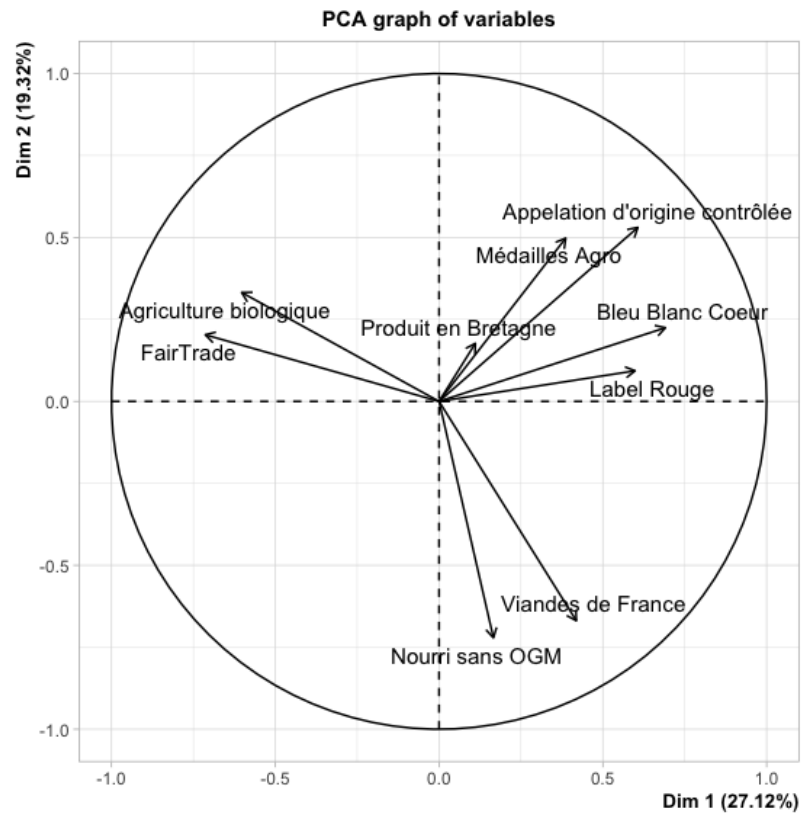
Does a product bearing this logo meet your quality expectations? Rate from 0 to 10.



Does a product bearing this logo meet your quality expectations?
Rate from 0 to 10.

	← →	📄	🔍 Filter							
	▲	↕ Agriculture biologique	↕ Label Rouge	↕ FairTrade	↕ Bleu Blanc Cœur	↕ Appellation d'origine contrôlée	↕ Produit en Bretagne	↕ Viandes de France	↕ Nourri sans OGM	↕ Médailles Agro
1		10	4.0	10	7	9.0	8.0	0	0	5
2		10	8.0	10	5	6.0	5.0	5	3	3
3		9	10.0	6	5	10.0	5.0	2	0	10
4		9	6.0	9	7	9.0	7.0	5	3	7
5		7	5.0	9	5	6.0	6.0	3	0	5
6		9	5.0	8	5	8.0	5.0	7	8	8
7		8	6.0	6	6	7.0	3.0	3	4	7
8		8	6.0	8	6	6.0	5.0	7	6	6
9		7	6.0	5	7	8.0	9.0	8	5	4
10		8	9.0	10	7	9.0	4.0	6	7	5
11		6	7.0	5	5	7.0	8.0	9	6	6
12		7	8.0	6	8	9.0	7.0	8	6	7
13		8	8.0	4	8	8.0	4.0	8	9	5
14		8	10.0	5	7	9.0	5.0	8	4	4
15		8	8.0	5	6	7.0	7.0	7	7	6

FactoMineR::PCA



Please NaileR, what
is the meaning of the
first dimension?



Please NaileR, what
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first dimension?



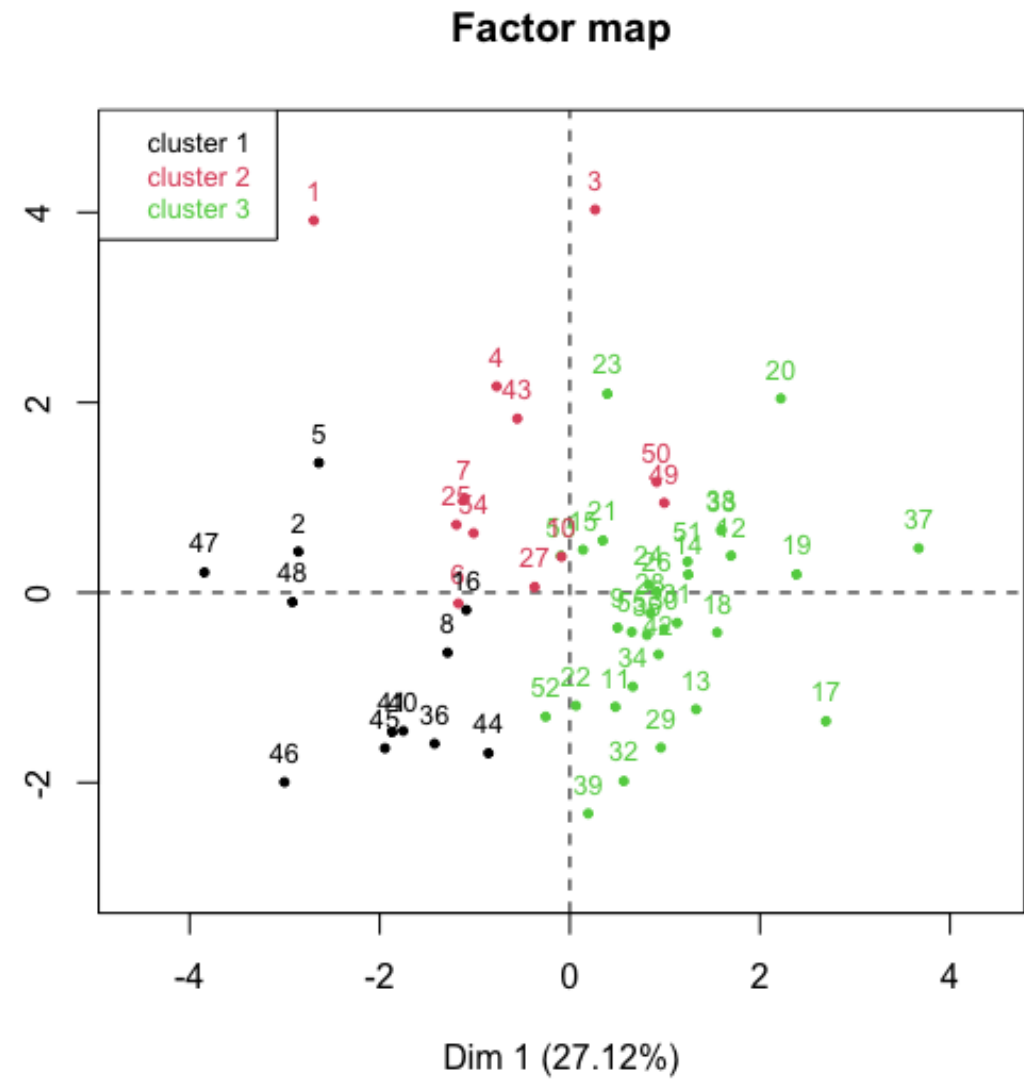
Based on the data, individuals on the **left side** of the scale tend to perceive French logos associated with organic or fair trade practices as having higher quality products, while those on the **right side** of the scale have the opposite opinion. This suggests that people on the left side are more likely to value **sustainable** and **socially responsible** food production methods, whereas those on the right side may be less concerned with these factors.

Please NaileR, what is the meaning of the first dimension?



I would name this scale the "Sustainability-Traditional Continuum". I chose this name because it captures the idea that individuals on one end of the scale prioritize sustainability (organic, fair trade) and quality, while those on the other end place more value on traditional practices and labels (such as appellation d'origine contrôlée). This continuum highlights the varying degrees to which people consider these factors when evaluating food quality.

FactoMineR::HCPC



Please NaileR, what
is the meaning of the
clusters?



Please NaileR, what is the meaning of the clusters?



- ****Group 1****: This group has higher ratings for FairTrade and Agriculture biologique. This suggests that participants in this group place importance on products that are sustainably sourced and environmentally friendly.
- ****Group 2****: This group has higher ratings for Agriculture biologique, Appelation d'origine contrôlée, FairTrade, and Médailles Agro. This indicates that participants in this group value high-quality, locally produced products with certifications or awards, possibly suggesting a focus on artisanal or small-scale farming practices.

Please NaileR,
what is the
meaning of the
clusters?

- ****Group 3****: This group has higher ratings for Bleu Blanc Coeur, Viandes de France, and Label Rouge. These labels are all associated with French agricultural standards and certifications, implying that participants in this group prioritize products that meet certain national quality standards.

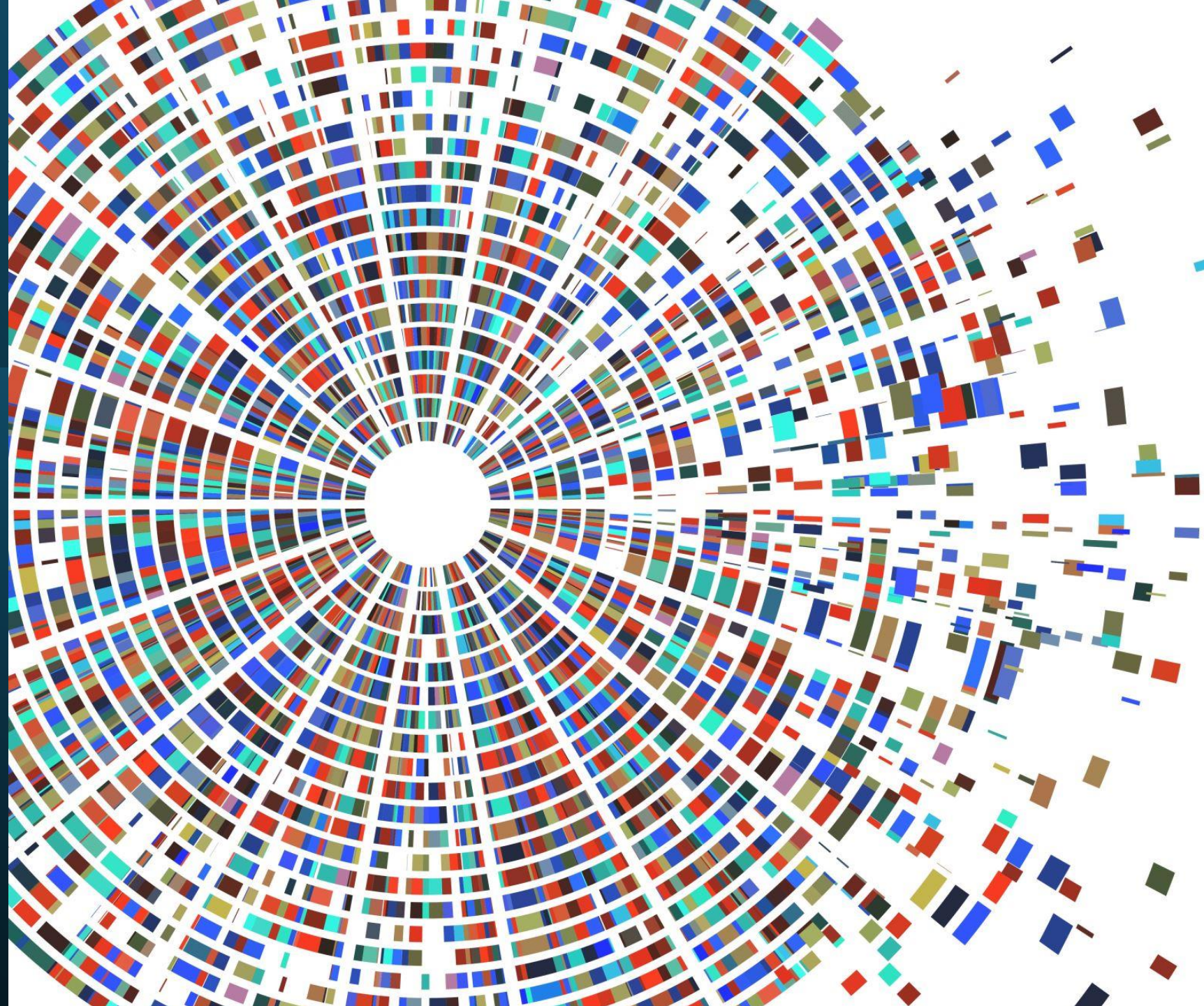


Please NaileR,
what is the
meaning of the
clusters?

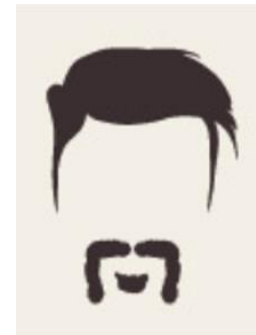
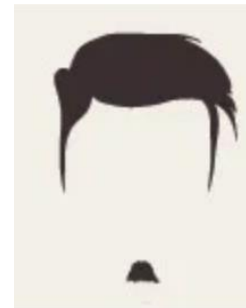
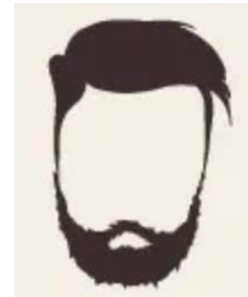
- Based on these conclusions, I would give each group a new name as follows:
 - * **Group 1**": "Sustainable Shoppers"
 - * **Group 2**": "Artisanal Enthusiasts"
 - * **Group 3**": "French Food Connoisseurs"



Analyzing a set of
stimuli/products
with NaileR



Dear **consumer**, in three words who do you think this person is?



##	self-centered	bobo	does not care	tanned	cowboy	edgy	musician
## B1	0	1	0	1	0	0	3
## B2	1	0	0	1	0	0	0
## B3	0	2	1	0	0	0	0
## B4	0	1	1	0	0	1	0
## B5	0	0	0	0	0	0	3
## B6	1	1	0	1	1	0	1
## B7	0	1	0	0	0	0	0
## B8	0	0	0	1	2	0	0

##	poorly	cared for	disorganized	adult	skinny	disturbed	suit	executive	ranch
## B1	0		0	1	0	0	1	2	0
## B2	0		0	0	0	0	0	0	0
## B3	1		0	0	0	0	0	0	0
## B4	0		0	0	0	0	0	0	0
## B5	0		0	0	2	0	0	0	0
## B6	0		1	0	2	0	0	1	0
## B7	0		0	0	3	1	0	0	0
## B8	0		0	1	0	0	0	0	1

The consumers data...

Please NaileR,
based on the
previous
contingency table,
could you **describe**,
the beards?





Description of the beards

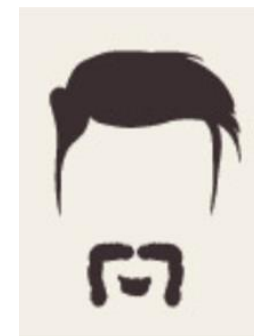
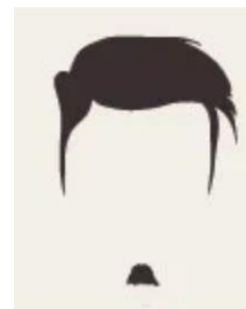
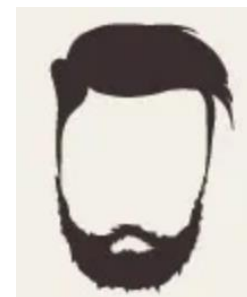


```
## Here are the names I came up with for each beard type, along with a summary of what makes them unique:
##
## **B1:
## This beard is characterized by its neatness, classic style, and modern elegance. It's perfect for the refined gentleman who wants to make a statement.
##
## **B2:
## This beard is all about being young, sporty, and seductive. It's ideal for the confident charmer who wants to turn heads.
##
## **B3:
## This beard combines elements of the lumberjack look with hipster flair. It's perfect for the strong, rugged individualist who wants to make a statement.
##
## **B4:
## This beard is characterized by its simplicity and lack of fuss. It's ideal for the busy dad who wants a low-maintenance beard that still looks great.
##
## **B5:
## This beard features a goatee and is perfect for the quirky, computer-savvy individual who wants to express their personality through their facial hair.
##
## **B6:
## This beard is characterized by its precision-cut style and pickiness. It's ideal for the discerning gentleman who wants a beard that's perfectly coiffed.
##
## **B7:
## This beard is all about being old-fashioned, shy, and a bit quirky. It's perfect for the introverted individual who wants to express their unique personality through their facial hair.
##
## **B8:
## This beard combines elements of the vintage biker look with a touch of Italian flair. It's ideal for the free-spirited individual who wants a beard that reflects their adventurous lifestyle.
```

Please NaileR,
based on the
previous
contingency table,
could you **name**,
the beards?



Write names on a piece of paper, the person who finds the most names wins...



Please NaileR, based on the previous contingency table, could you **name** the beards?

The Elegant Scholar



The Young Charmer



The Lumberjack Hipster



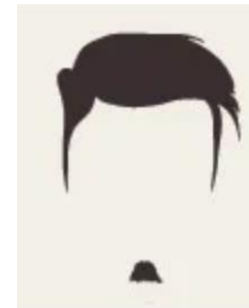
The Busy Dad



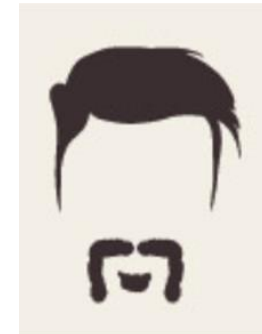
The Goatee-Sporting Geek



The Picky Gent



The Shy Old Soul



The Vintage Biker

Thank you for your
attention!

